

Varian Medical Systems Eclipse

ESAPI v15.6 / v16.1

Production Ready (v3.0.0)

EclipseDataMiner

Version 3.0.0 (Released: September 26, 2026)

Radiation Therapy Plan Data Mining & Quality Assurance Technical Manual

This manual provides comprehensive technical documentation for **EclipseDataMiner**, a high-throughput, memory-safe data mining and quality management system querying 10,000+ radiation therapy plans across Varian Eclipse (ESAPI). It covers system installation, operational workflows, structure nomenclature mapping (Pre-Scan & Alias Mapping), Dose-Quality Parameter (DQP) extraction, delivery modulation complexity modeling (MCS, Edge Metric, Leaf Travel Length, Arc Length), STA worker streaming architecture, and clinical commissioning procedures.

TARGET ENVIRONMENT Varian Eclipse v15.6 / v16.1 (ESAPI)	EXECUTION FRAMEWORK Microsoft .NET Framework 4.6.1 (x64)
DEPLOYMENT ARCHITECTURE Single Standalone Executable (Costura.Fody)	VALIDATION STATUS 116/116 Tests Passed (100% PASS)

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Chapter 1

System Overview & Quick Start

Objective of EclipseDataMiner, core features, UI layout, and deployment procedures

A standalone **10,000-plan-scale clinical data mining application** running on Varian Eclipse (ESAPI).

It cross-searches patient databases for treatment plans (PlanSetup / PlanSum) matching complex clinical criteria, extracts dose-quality parameters (DQP), delivery parameters, clinical protocols, optimization objectives, and modulation complexity metrics (Modulation Complexity Score, Edge Metric, etc.) with high throughput and memory safety, and streams results into normalized CSV and machine-learning-ready JSON Lines (`.jsonl`) files.

☀️ Key Features (v3.0.0)

- 🌐 **Standard English UI & Bilingual Documentation:**
 - Clean, international-standard English UI designed for radiation oncology clinical workflows.
 - Comprehensive technical documentation and user manuals provided in both English and Japanese.
- 🚀 **10,000-Scale High-Throughput Streaming Pipeline:**
 - Immediately flushes extracted data to disk per plan/patient without accumulating in memory, maintaining a flat memory footprint (~few tens of MBs) even during long runs.
- 🔍 **Structure Pre-Scan & Multi-Rule Alias Mapping:**
 - High-speed pre-scan of structure IDs across matching plans, aggregating nomenclature variations (`PTV_60` , `ptv60` , `PTV-60Gy` , etc.) into unified columns via Exact, Substring, or Regex matching.
 - Reusable JSON rule persistence.
- ⚙️ **Hierarchical AND / OR Filtering & Dose Presence Logic:**
 - Comma-delimited intra-field OR search combined with top-level AND / OR toggle switch.
 - Dose status filter (`ALL` , `HasDose` [calculated], `NoDose` [uncalculated]).
 - Numeric range expressions (`70-80`) and inequality filters (`≥ 10`) for doses and fraction counts.
 - Delivery parameter filters (Machine, Energy, Technique) and date range filtering.
 - Support for composite plans (`PLanSum`).
- 📺 **Simultaneous JSON Lines (`.jsonl`) Streaming for AI / ML:**
 - Preserves hierarchical plan data (beams, optimization constraints, DVH metrics) ready for direct ingestion in Python (Pandas, PyTorch, TensorFlow).
- 🛡️ **Privacy Protection & Patient De-Identification:**
 - SHA-256 patient ID hashing and automatic `REDACTED` masking of personal data (birth dates, approver names) for collaborative and external clinical research.

-  **Automatic Dose Unit Standardization (Unified Gy):**
 - Automatically identifies ESAPI `cGy` / `Gy` inconsistencies and standardizes all values to `Gy`.
-  **Robust Dedicated STA Worker Thread & Memory Safety:**
 - Background STA thread prevents UI freezing during intensive ESAPI queries.
 - Periodic `GC.Collect()` every 200 patients and strict `try-finally` object disposal to eliminate memory leaks.
-  **Single Standalone Executable (Costura.Fody):**
 - All external NuGet dependencies are bundled inside `EclipseDataMiner.exe`. Ready to run on any clinical Eclipse workstation without installer or DLL deployment.

System Requirements

- **Operating System:** Windows 10 / 11 / Windows Server (64-bit)
- **Treatment Planning System:** Varian Eclipse v15.6 / v16.1
- **Runtime:** .NET Framework 4.6.1 or higher
- **Permissions:** ESAPI script execution permissions (Run on clinical workstation or TBox)

Quick Start & Installation

This tool is distributed as a single standalone executable.

1. Obtain the Executable:

- Download `release\EclipseDataMiner_v3.0.0.exe` (or build `EclipseDataMiner\bin\x64\Release\EclipseDataMiner.exe`).

2. Deploy to Eclipse Workstation:

- Copy `EclipseDataMiner_v3.0.0.exe` to any folder on your Eclipse workstation.

3. Launch:

- Double-click `EclipseDataMiner_v3.0.0.exe` to launch.
-

User Interface Overview

The screenshot displays the 'Plan Search' tab of the EclipseDataMiner v3.0.0 application. The interface is divided into four sequential tabs: 1- Plan Search, 2- Structure Mapping, 3- Dose Quality Parameters (DQP), and 4- Extraction & Analysis Options.

Plan Search & Filter Criteria:

- Preset:** Prostate VMAT 78Gy (Standard) [Save] [Delete] [Regex Hints]
- Description:** Prostate VMAT 78Gy (Standard)
- Patient ID:** 3315206, 3315207, 3315308, 3315500 [Contains]
- Course ID:** [Contains]
- Plan ID:** [Contains]
- Target Volume:** [Contains]
- Dose/Fr [Gy]:** 78
- Fractions:** 5
- Total Dose [Gy]:** 78
- Logic:** AND (selected) OR
- Unapproved:** [checked] **Plan approved:** [checked] **TRT approved:** [checked]
- Dose:** HasDose [Include PlanSum]
- Advanced Filters:** (Machine, Energy, Technique, Date Range)

Searches treatment plans matching specified criteria across the database and previews results in the table below. [Search Plans]

Select All **Unselect All** **Invert** **6 / 6 plans selected**

Extract	Patient ID	Course ID	Plan ID	Target Volume	Type	Approval Status	Dose	Dose/Fr	Fractions	Total Dose	Machine	Energy	Tec
☒	3315206	0301	918	-	PlanSetup	TreatmentApproved	✓	78.00 Gy	5	78.00 Gy	TMAT 5	6X	VMA
☒	3315207	0302	919	-	PlanSetup	TreatmentApproved	✓	78.00 Gy	5	78.00 Gy	TMAT 6	6X	VMA
☒	3315308	0302	2900	-	PlanSetup	TreatmentApproved	✓	78.00 Gy	5	78.00 Gy	TMAT 5	10X	VMA
☒	3315500	0304	2901	-	PlanSetup	TreatmentApproved	✓	78.00 Gy	5	78.00 Gy	TMAT 6	6X	VMA
☒	3315501	0403	2902	-	PlanSetup	TreatmentApproved	✓	78.00 Gy	5	78.00 Gy	TMAT 6	6X	VMA
☒	3315522	0403	7003	-	PlanSetup	Completed	—	78.00 Gy	5	78.00 Gy	TMAT 8	10X	VMA

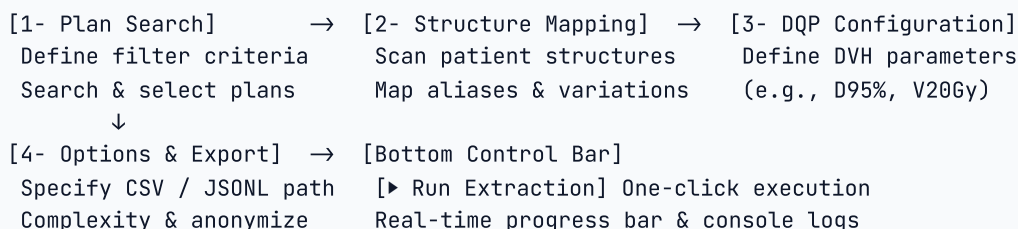
Console Output [Copy] [Clear]

```
Ready to extract.
[INFO] Loaded preset: Prostate VMAT 78Gy (Standard)
[INFO] Matched 6 plans matching clinical criteria across 5 patients.
[INFO] Structure Mapping verified (PTV, Rectum, Bladder, FemoralHeads).
[INFO] Ready for execution.
```

Run Extraction [Cancel] **Idle / Ready (6 plans selected for extraction)** **100%**

Operational Workflow

The application UI consists of 4 sequential tabs, guiding users intuitively from left to right:



1. Tab 1: 1- Plan Search:

- Configure search criteria (Patient ID, Plan ID, Dose status, Approval status, Machine, Dates) or load a saved Preset.
- Click "🔍 Search Plans" to scan the database, then select target plans in the results data grid.

2. Tab 2: 2- Structure Mapping:

- Click " **Pre-Scan Structures**" to discover structure names across target plans and consolidate variations into unified aliases.

3. **Tab 3: 3- Dose Quality Parameters (DQP):**

- Configure dose/volume evaluation metrics (e.g., D95%, V20Gy, Mean, Max). Supports CSV import/export.

4. **Tab 4: 4- Extraction & Analysis Options:**

- Choose output CSV destination path ( / ).
- Enable Beam Parameters, Plan Complexity (MCS, Edge Metric), Anonymization, and JSONL output options.

5. **Execute Extraction (Bottom Bar):**

- Click " **Run Extraction**" to start the extraction pipeline. Monitor live progress and execution logs in the bottom status panel.

Export Data Specifications

1. **CSV Output (Flat Format)**

- One row per plan (comma-separated, UTF-8 with BOM).
- Structured in 4 sections: Core columns (11), Optional metadata/parameters (up to 10), Baseline structure statistics (4 per structure), and Dynamic DQP metrics. For a complete column-by-column reference, see [User Manual Section 4 \(docs/MANUAL.en.md#4\)](#).
- Newline characters are replaced with spaces; commas and quotes are escaped properly.
- Missing or non-applicable values are exported as  .

Column Name (Examples)	Description	Example
Patient ID	Patient identifier (SHA-256 hash if anonymized)	PT_1001 or e3b0c442...
Course ID / Plan ID	Course and Plan IDs	C1 , VMAT_Prostate
DosePerFraction[Gy]	Prescription dose per fraction (Gy)	2.00
TotalDose[Gy]	Total prescription dose (Gy)	60.00
ApprovalStatus	Clinical approval status	TreatmentApproved
MU	Beam MU values (semicolon-separated)	B1:145.2;B2:138.6
Machine/Energy/Tech/PlanType	Delivery parameters summary	TrueBeam:6X:ARC:VMAT
PlanComplexity	Modulation metrics (Beam:MCS,EM,LeafTravel,ArcLength)	(B1:0.28,0.12,1850.4,360.0)
<Structure>—Volume[cc]	Structure volume (auto-exported)	42.50
<Structure>—Max dose[Gy]	Structure maximum dose (auto-exported)	64.20
<Structure>—Mean dose[Gy]	Structure mean dose (auto-exported)	61.80
<Structure>—Min dose[Gy]	Structure minimum dose (auto-exported)	58.10
<Structure>—D95%[Gy]	Calculated DQP dose value	59.80
<Structure>—V70%[%]	Calculated DQP relative volume value	18.50

2. JSON Lines (.jsonl) Output

- One complete JSON object per line.
- Easily loadable into Python:

```
import json
with open("DataMiningOutput.csv.jsonl", "r", encoding="utf-8") as f:
    plans = [json.loads(line) for line in f]
print(f"Loaded {len(plans)} plans.")
```

Technical Documentation Suite

Document	Description
 User Manual	English (docs/MANUAL.en.md) 日本語 (docs/MANUAL.md) — Complete step-by-step UI guide, DQP templates, Python integration examples
 Architecture Specification	English (docs/ARCHITECTURE.en.md) 日本語 (docs/ARCHITECTURE.md) — Layered architecture, STA worker model, memory management, and sequence diagrams
 Clinical Commissioning Protocol	English (docs/COMMISSIONING.en.md) 日本語 (docs/COMMISSIONING.md) — TG-275-compliant clinical commissioning guide, validation checklists, approval forms
 Troubleshooting & FAQ	English (docs/TROUBLESHOOTING.en.md) 日本語 (docs/TROUBLESHOOTING.md) — Common issues, error resolutions (file locks, memory, ESAPI connection), FAQ
 Contributing Guidelines	English (docs/CONTRIBUTING.en.md) 日本語 (docs/CONTRIBUTING.md) — Environment setup, build/test pipelines, coding conventions, git workflow
 Design Specification	English (docs/DESIGN_SPECIFICATION.en.md) 日本語 (docs/DESIGN_SPECIFICATION.md) — Comprehensive software specifications and ESAPI control specifications
 Standard Development Plan	English (docs/STANDARD_DEVELOPMENT_PLAN.en.md) 日本語 (docs/STANDARD_DEVELOPMENT_PLAN.md) — 7 Core Quality Principles, 4-tier DoD gates, and reproducibility protocols
 Changelog	English (docs/CHANGELOG.en.md) 日本語 (docs/CHANGELOG.md) — Version-by-version release history (Keep a Changelog standard)
 Third-Party Notices	THIRD-PARTY-NOTICES.md — Open-source software licenses and copyright acknowledgments
 Consolidated Technical Manual PDF	English (release/EclipseDataMiner_v3.0.0_Manual.en.pdf) 日本語 (release/EclipseDataMiner_v3.0.0_Manual.pdf) — Unified publication-quality manual formatted for A4 printing

Testing & Quality Assurance

This repository includes a multi-layered automated test suite using `EclipseDataMiner.Tests` (MSTest), verifying all business and numerical logic without requiring an active ESAPI database connection.

- **Total Unit Tests:** 116 tests (100% PASS)
- **Coverage:** Dose normalization (`ToGy()`), string sanitization, patient de-identification, streaming CSV/JSONL pipelines, hierarchical AND/OR search logic with UI synchronization, dose presence filter (`HasDose` / `NoDose`), inequality/range parsing, beam/date filters, preset persistence, 2-pane structure mapping with real-time preview, rule priority resolution (Exact > Contains > Regex), complexity algorithms benchmarked against published literature (MCS, Edge Metric, Leaf Travel Length, Arc Length), XAML resource integrity, and UI screenshot generation.

 License

This software is released under the [MIT License](#). For third-party open-source components and licenses included in this project, please refer to [THIRD-PARTY-NOTICES.md](#).

Chapter 2

Clinical Operational Manual & Features

Four main tabs workflow, plan search & filters, structure mapping, DQP metrics, and extraction execution

This document is the official operational guide and comprehensive technical reference for **EclipseDataMiner v3.0.0**.

Note: The user interface (UI) of this application is displayed in English by default. This manual references all UI components using their exact English labels.

1. Overview and UI Layout

EclipseDataMiner is a standalone WPF application that cross-searches the Varian Eclipse (ESAPI) database for treatment plans matching complex clinical criteria, and safely extracts Dose-Quality Parameters (DQP), delivery parameters, and plan modulation complexity metrics (MCS, Edge Metric, Leaf Travel Length, Arc Length) with high throughput and low memory usage.

The main window consists of **4 numbered tabs at the top**, a **dark terminal console in the middle**, and an **execution control bar with a progress bar at the bottom**:

```
+-----+
| [1- Plan Search] [2- Structure Mapping] [3- DQP] [4- Options] |
| | | | |
| << Tab 1: 1- Plan Search >> |
| - CARD: Plan Search & Filter Criteria |
|   - Patient / Course / Plan / Target / Dose / Status / Advanced |
|   - [ Quick Scan Database and Preview Plans [Search Plans] ] |
| - TABLE: Matched Plans (Full Metadata & Delivery Dates) |
|   - [Select All] [Unselect All] [Invert] (Select Plans to Extract)|
| | | | |
| << Tab 4: 4- Extraction & Analysis Options >> |
| - CARD: Export Destination ( CSV Path ... [Browse] [Open Folder] )|
| - CARD: Metadata / Complexity / Output & Privacy Formats |
+-----+
| CARD: Console Output [Copy] [Clear] |
| [ Dark Slate Terminal Console (Auto-scroll, Consolas font) ] |
+-----+
| [▶ Run Extraction] [■ Cancel] [===== 42% =====] |
| Status: Processing patient 42/100 (Extracted plans: 84)... |
+-----+
```

1- Plan Search

2- Structure Mapping

3- Dose Quality Parameters (DQP)

4- Extraction & Analysis Options

Plan Search & Filter Criteria

Preset: Prostate VMAT 78Gy (Standard)

Save

Delete

Regex Hints

Description: Prostate VMAT 78Gy (Standard)

Patient ID: 3315206, 3315207, 3315308, 3315500

Contains

Course ID:

Contains

Logic: AND OR

Plan ID:

Contains

Target Volume:

Contains

Unapproved

Plan approved

TRT approved

Dose/Fr [Gy]: 78

Fractions: 5

Total Dose [Gy]: 78

Dose: HasDose

Include PlanSum

Advanced Filters (Machine, Energy, Technique, Date Range)

Searches treatment plans matching specified criteria across the database and previews results in the table below.

Search Plans

Select All

Unselect All

Invert

6 / 6 plans selected

Extract	Patient ID	Course ID	Plan ID	Target Volume	Type	Approval Status	Dose	Dose/Fr	Fractions	Total Dose	Machine	Energy	Ted
☒	3315206	0301	918	-	PlanSetup	TreatmentApproved	✓	78.00 Gy	5	78.00 Gy	TMAT 5	6X	VMA
☒	3315207	0302	919	-	PlanSetup	TreatmentApproved	✓	78.00 Gy	5	78.00 Gy	TMAT 6	6X	VMA
☒	3315308	0302	2900	-	PlanSetup	TreatmentApproved	✓	78.00 Gy	5	78.00 Gy	TMAT 5	10X	VMA
☒	3315500	0304	2901	-	PlanSetup	TreatmentApproved	✓	78.00 Gy	5	78.00 Gy	TMAT 6	6X	VMA
☒	3315501	0403	2902	-	PlanSetup	TreatmentApproved	✓	78.00 Gy	5	78.00 Gy	TMAT 6	6X	VMA
☒	3315522	0403	7003	-	PlanSetup	Completed	—	78.00 Gy	5	78.00 Gy	TMAT 8	10X	VMA

Console Output

Copy

Clear

Ready to extract.
[INFO] Loaded preset: Prostate VMAT 78Gy (Standard)
[INFO] Matched 6 plans matching clinical criteria across 5 patients.
[INFO] Structure Mapping verified (PTV, Rectum, Bladder, FemoralHeads).
[INFO] Ready for execution.

Run Extraction

Cancel

Idle / Ready (6 plans selected for extraction)

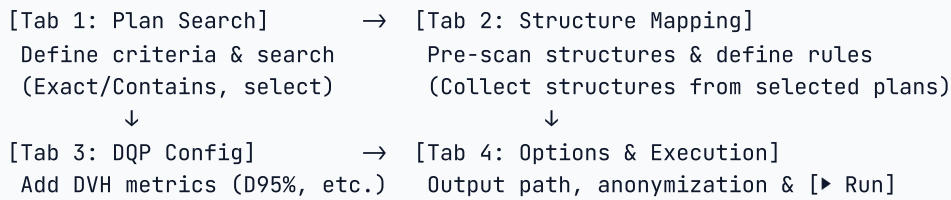
100%

2. Four Main Tabs Overview

Tab Number & Name	Key Function
1- Plan Search	Define search filters (Patient ID, Plan ID, Prescription Dose, Approval Status, Machine, Date Range) and manage presets. Rapidly scan the database and select target plans via checkboxes.
2- Structure Mapping	High-speed pre-scan of structure nomenclatures across target plans (PTV_60 , ptv60 , PTV-60Gy , etc.) and consolidate variations into unified Target Aliases.
3- Dose Quality Parameters (DQP)	Configure dose/volume evaluation metrics (e.g., D95%, V20Gy, Mean, Max) with CSV import/export. Note: Basic statistical parameters (Volume, Min, Mean, Max) are automatically exported without manual registration.
4- Extraction & Analysis Options	Output CSV file path, delivery parameters, plan complexity metrics (MCS / Edge Metric), SHA-256 patient de-identification, and JSONL streaming options.

3. Step-by-Step Workflow

The standard operational workflow proceeds sequentially through the numbered tabs (**Tab 1 → Tab 2 → Tab 3 → Tab 4 → Bottom Execution Bar**):



Step 1: Define Plan Search Criteria & Select Plans (Tab 1: 1- Plan Search)

Plan Search & Filter Criteria

Preset: **Prostate VMAT 78Gy (Standard)** [Save] [Delete] [Regex Hints]

Description: Prostate VMAT 78Gy (Standard)

Patient ID: 3315206, 3315207, 3315308, 3315500 [Contains] Course ID: [Contains] Logic: ☒ AND ☐ OR

Plan ID: [Contains] Target Volume: [Contains] ☒ Unapproved ☒ Plan approved ☒ TRT approved

Dose/Fr [Gy]: 78 Fractions: 5 Total Dose [Gy]: 78 Dose: HasDose ☐ Include PlanSum

Advanced Filters (Machine, Energy, Technique, Date Range)

Searches treatment plans matching specified criteria across the database and previews results in the table below. [Search Plans]

Select All Unselect All Invert 6 / 6 plans selected

Extract	Patient ID	Course ID	Plan ID	Target Volume	Type	Approval Status	Dose	Dose/Fr	Fractions	Total Dose	Machine	Energy	Tech
☒	3315206	0301	918	-	PlanSetup	TreatmentApproved	✓	78.00 Gy	5	78.00 Gy	TMAT 5	6X	VMA
☒	3315207	0302	919	-	PlanSetup	TreatmentApproved	✓	78.00 Gy	5	78.00 Gy	TMAT 6	6X	VMA
☒	3315308	0302	2900	-	PlanSetup	TreatmentApproved	✓	78.00 Gy	5	78.00 Gy	TMAT 5	10X	VMA
☒	3315500	0304	2901	-	PlanSetup	TreatmentApproved	✓	78.00 Gy	5	78.00 Gy	TMAT 6	6X	VMA
☒	3315501	0403	2902	-	PlanSetup	TreatmentApproved	✓	78.00 Gy	5	78.00 Gy	TMAT 6	6X	VMA
☒	3315522	0403	7003	-	PlanSetup	Completed	—	78.00 Gy	5	78.00 Gy	TMAT 8	10X	VMA

Console Output [Copy] [Clear]

```

Ready to extract.
[INFO] Loaded preset: Prostate VMAT 78Gy (Standard)
[INFO] Matched 6 plans matching clinical criteria across 5 patients.
[INFO] Structure Mapping verified (PTV, Rectum, Bladder, FemoralHeads).
[INFO] Ready for execution.
  
```

[▶ Run Extraction] [Cancel] Idle / Ready (6 plans selected for extraction) 100%

1. Search & Filter Criteria:

◦ Search Presets (Presets) & Protocol Descriptions:

- The " **Preset:**" bar at the top of the card allows you to save, load, and delete clinical search criteria sets with a single click.

- **In-place Description Editing:** When a preset is selected, its description is displayed in an editable text box. You can annotate clinical protocols or trial numbers directly and click " Save" to persist both criteria and descriptions.
- **Storage Location (Presets/ Folder):** Presets are stored as individual JSON files in the Presets/ folder alongside the application (e.g., Presets/Prostate VMAT 78Gy.json). This enables simple backup and sharing between workstations. Pre-configured samples are provided in Templates/Presets/ .
- **Patient ID / Course ID / Plan ID / Target Volume:**
 - Supports comma-separated multiple OR searches (e.g., 1001, 1002, 2005 or VMAT, IMRT).
 - **NOT / Exclusion Filters (! or -):** Prefixing a keyword with ! or - excludes plans matching that pattern (e.g., VMAT, IMRT, !QA, !Test).
 - **Match Mode Selection (Contains / Exact / Regex):**
 - Contains (Default): Substring matching, case-insensitive.
 - Exact : Strict exact match. Specifying Exact with Plan matches only Plan , excluding Plan1 or MyPlan .
 - Regex : Full regular expression matching (e.g., ^Plan_\d+\$ or (VMAT|IMRT)_Prostate).
 - **Regex Cheat Sheet (? Regex Hints):**
 - Click to open clinical regex patterns for Plan ID and Patient ID. Click "Insert" to automatically insert the pattern and switch mode to Regex.
- **Prescription Dose & Fraction Criteria (Dose/Fr, Fractions, Total Dose):**
 - Supports exact values (78 , 2.0), numeric ranges (70-80 or 70 ~ 80 , fractions 33-35), and inequalities (≥ 10 , < 30 , ≤ 5).
 - Plans with uncalculated doses or NaN / Infinity are strictly excluded.
- **Dose Calculation Status (Dose Presence: All / HasDose / NoDose):**
 - All : All plans regardless of dose calculation status (default).
 - HasDose : Matches **only plans with calculated 3D dose**.
 - NoDose : Matches **only uncalculated plans** (e.g., preliminary contouring or beam setup plans).
- **Global Logic Switch (AND / OR):**
 - AND : Plan must satisfy **all** specified criteria.
 - OR : Plan matches if **any single** criterion is satisfied.
- **Approval Status / Include PlanSum:**
 - Checkboxes for Unapproved , Plan approved , and TRT approved .
 - Check Include PlanSum to include composite plan sums in search results.
- **Advanced Filters (Machine, Energy, Technique, Date Range):**
 - Expand ▼ Advanced Filters to filter by delivery equipment and dates:
 - Machine ID (e.g., TrueBeam, Clinac_iX, !QA_Linac)
 - Energy (e.g., 6X, 10X, !6FFF)
 - Technique (e.g., ARC, STATIC, !SRS)

- **Date Target** (**TreatmentApprovalDate** , **PlanningApprovalDate** , **CreationDate**) and **Date Range** (**From** ~ **To**).

2. Execute Plan Search (**Search Plans**):

- Click " **Search Plans**" in the blue footer bar. Matching plans are scanned from the ESAPI database and listed in the table below.
- The table summarizes aggregated beam parameters (Machine, Energy, Technique), delivery dates, and dose calculation markers (Dose column).

3. Select Plans in the Matched Plans DataGrid (**Extract** **Column Checkboxes**):

- Use the **Extract** checkboxes in the first column to select or exclude individual plans.
- Toolbar buttons: "**Select All**", "**Unselect All**", and "**Invert**" allow batch toggling.
- **Selection Propagation:**
 - **Checked plans** (**Extract = true**) are directly propagated to **Step 2 (Structure Mapping Pre-Scan)** and **Step 5 (Run Extraction)**.
 - This prevents wasteful full-database queries and pinpoints processing only to the selected plans.

Step 2: Structure Pre-Scan & Nomenclature Mapping (Tab 2: 2- **Structure Mapping**)

Consolidate institutional naming variations (e.g., **PTV_60** , **ptv60** , **PTV-60Gy**) into unified column headers (Target Alias). The tab features a 2-pane layout separated by a draggable GridSplitter.

1- Plan Search 2- Structure Mapping 3- Dose Quality Parameters (DQP) 4- Extraction & Analysis Options

Mapping Rules (Rule Definitions)

Extract	Structure ID / Pattern	Match Mode	Target Alias
☒	PTV_78Gy	Exact	PTV
☒	PTV	Contains	PTV
☒	(?i)^rectum.*	Regex	Rectum
☒	(?i)^bladder.*	Regex	Bladder
☒	Femur_R	Contains	FemoralHead_R
☒	Femur_L	Contains	FemoralHead_L

Discovered Structures & Mapping Preview

Raw Structure ID	Hits	Resolved Alias	Status
PTV_78Gy	6	PTV	Mapped (Exact) [Merged: 2]
ptv78	4	PTV	Mapped (Contains) [Merge]
Rectum	6	Rectum	Mapped (Regex)
Bladder	6	Bladder	Mapped (Regex)
Femur_R	5	FemoralHead_R	Mapped (Contains)
Femur_L	5	FemoralHead_L	Mapped (Contains)
PenileBulb	3	PenileBulb	Unmapped (Raw)
Bowel	2	Bowel	Unmapped (Raw)

Console Output

```
Ready to extract.
[INFO] Loaded preset: Prostate VMAT 78Gy (Standard)
[INFO] Matched 6 plans matching clinical criteria across 5 patients.
[INFO] Structure Mapping verified (PTV, Rectum, Bladder, FemoralHeads).
[INFO] Ready for execution.
```

Run Extraction **Cancel** Idle / Ready (6 plans selected for extraction) 100%

1. Open **Tab 2: 2- Structure Mapping**.

2. **Right Pane: Execute Structure Pre-Scan** (🔍 Pre-Scan):

- Click "🔍 Pre-Scan" (Primary Button).
- **Automatic Scope Detection:**
 - Scans structures **only within plans checked in Tab 1** (**Extract = true**).
 - The scope badge on the toolbar (🌐 Target: 3 / 3 Selected Plans) displays the target count in real time.
 - If search was not executed in Tab 1, all plans matching criteria are scanned (🌐 Target: ALL Criteria Matching Plans).
 - **Executing pre-scan does not overwrite or clear rules in the left pane.**
- Upon scan completion, all discovered Structure IDs, occurrence counts (Count), resolved aliases, and match status are displayed.

3. **Left Pane: Mapping Rule Management:**

- **Load JSON / Save JSON:** Load pre-configured mapping dictionaries (e.g., Templates/StructureMapping_Prostate.json) and save edited dictionaries.
- **+ Add Rule / - Delete Selected:** Add or delete rules (supports continuous deletion via the Delete key).

- **▲ Up / ▼ Down:** Adjust rule evaluation order (supports keyboard shortcuts **Alt+Up/Down** and **Ctrl+Up/Down**).
- **Target Alias / Match Mode:**
 - Choose **Exact** (strict match), **Contains** (substring), or **Regex** (regular expression).
 - **Specificity Priority** (**Exact** > **Contains** > **Regex**): Exact matches take precedence, followed by Contains, and lastly Regex. This allows comprehensive regex rules and specific exceptions to coexist safely.
 - **Real-time Syntax Validation:** Invalid regex patterns highlight the cell with a red border and display tooltip error messages.
 - **? Regex Hints ▼** : Insert common clinical regular expression snippets with one click.
- **Extract Checkbox:** Uncheck structures that should be excluded from extraction (e.g., Couch, Marker).

4. Inspect Preview & Add Discovered Items:

- Filter structures in the right pane using quick search, status filters (**All** / **Unmapped Only** / **Mapped Only** / **Excluded Only**), or click "**Refresh**" to re-evaluate rules immediately.
- **Double-click** any structure row or click "**+ Add to Rules**" to immediately add it as a new rule in the left pane and refresh the preview.
- When multiple raw structures map to the same Target Alias, the status annotates **[Merged: N]**.

5. Bypass Structure Pre-Scan:

- If using an established rule dictionary, check "**Bypass Structure Pre-Scan**" in Tab 4 to skip the pre-scan confirmation prompt.

Step 3: Dose Quality Parameters (DQP) Configuration (Tab 3: 3- Dose Quality Parameters)

1-  Plan Search 2-  Structure Mapping 3-  Dose Quality Parameters (DQP) 4-  Extraction & Analysis Options

 **Auto Export of Basic Statistics:** Volume[cc], Max dose[Gy], Mean dose[Gy], and Min dose[Gy] are automatically exported for all mapped structures, so individual registration is not required.

+ Add DQP - Delete ▲ Up ▼ Down  Load CSV  Save CSV

Structure Name / Target Alias	DQP Type	Value	Input Unit	Output Unit
PTV	D (Dose at Volume)	95	Relative [%]	Relative [%]
PTV	D (Dose at Volume)	98	Relative [%]	Relative [%]
PTV	D (Dose at Volume)	2	Relative [%]	Relative [%]
Rectum	V (Volume at Dose)	70	Absolute [Gy / cc]	Relative [%]
Rectum	V (Volume at Dose)	50	Absolute [Gy / cc]	Relative [%]
Rectum	V (Volume at Dose)	40	Absolute [Gy / cc]	Relative [%]
Bladder	V (Volume at Dose)	70	Absolute [Gy / cc]	Relative [%]
Bladder	V (Volume at Dose)	65	Absolute [Gy / cc]	Relative [%]
FemoralHead_R	V (Volume at Dose)	50	Absolute [Gy / cc]	Relative [%]
FemoralHead_L	V (Volume at Dose)	50	Absolute [Gy / cc]	Relative [%]

Console Output  

```
Ready to extract.
[INFO] Loaded preset: Prostate WAT 78Gy (Standard)
[INFO] Matched 6 plans matching clinical criteria across 5 patients.
[INFO] Structure Mapping verified (PTV, Rectum, Bladder, FemoralHeads).
[INFO] Ready for execution.
```

 Run Extraction  Cancel Idle / Ready (6 plans selected for extraction) 100%

1. Open **Tab 3: 3-  Dose Quality Parameters (DQP)**.

2. **Automatic Baseline DVH Statistics & Normalization Reference:**

- **Baseline statistics** (`Volume[cc]` , `Max dose[Gy]` , `Mean dose[Gy]` , `Min dose[Gy]`) are **automatically exported for all mapped structures**. You do not need to register them individually.
- **Standard Plan** (`PlanSetup`) **Relative Dose Normalization:**
 - Relative dose (%) is normalized against the plan's **Total Prescription Dose** (`TotalDose`).
- **Composite Plan** (`PlanSum`) **Relative Dose Calculation Behavior:**
 - `PlanSum` is a composite of multiple treatment plans, and by ESAPI specification, does not possess a single `TotalDose` prescription dose.
 - **Relative Input Dose** (`InputUnit = Relative [%]` , e.g., `V70%`): Falls back to using the **structure's Maximum Dose** (`Max dose`) as the 100% reference dose (e.g., `V50%` calculates volume receiving 50% of the structure's maximum dose).
 - **Relative Output Dose** (`OutputUnit = Relative [%]` , e.g., `D95[%]`): If composite normalization is unconfigured in Eclipse, outputs `N/A` safely.
 - **Relative Volume (%)**: Evaluated against the physical structure volume and computes normally on PlanSums (e.g., `D95[Gy]` , `V50Gy[%]`).

- **Clinical Recommendation:** For cohorts including PlanSum, it is strongly recommended to specify dose criteria in **Absolute Dose [Gy]** (e.g., D95%[Gy], V50Gy[%], D0.1cc[Gy]) to ensure consistent dosimetric evaluation.

3. Add, Edit, and Delete DQP Metrics:

- **+ Add DQP:** Add a new evaluation metric row.
- **- Delete Selected:** Remove selected row (supports continuous Delete key deletion).
- **▲ Up / ▼ Down:** Reorder evaluation metrics (supports keyboard shortcuts Alt+Up/Down and Ctrl+Up/Down).
- **Structure Name / Alias:** Target structure name or Target Alias defined in Step 2.
- **DQP Type:**
 - **D (Dose at Volume)**: Dose at a specific volume (e.g., D95%, D0.1cc).
 - **V (Volume at Dose)**: Volume receiving a specific dose (e.g., V20Gy, V70%).
 - **DC (Dose Complement)**: Dose complement.
 - **CV (Complement Volume)**: Complement volume.
- **Value / Input Unit / Output Unit:** Set numeric value, input unit (Relative [%] / Absolute [Gy / cc]), and output unit.

4.  **Load CSV** /  **Save CSV:** Save and load protocol templates in CSV format (Header: structureName,DQPtype,DQPvalue,InputUnit,OutputUnit).

Step 4: Output Destination & Analysis Options (Tab 4: 4- ⚙️ Extraction & Analysis Options)

1- Plan Search 2- Structure Mapping 3- Dose Quality Parameters (DQP) 4- ⚙️ Extraction & Analysis Options

Export File Destination

Output CSV File: C:\EclipseDataMiner\Exports\Prostate_VMAT_Cohort_2026.csv Browse... Open Folder

Plan Metadata Extraction

☒ Planning Approver ☒ Planning Approval Date ☒ Calculation Model ☒ Normalization Mode

☒ Clinical Protocol ☒ Optimization Objectives

Plan Complexity Analysis

☒ Calculate Modulation Complexity Score (MCS), Edge Metric, Leaf Travel, Arc Length
* Modulation complexity metrics are automatically calculated from MLC leaf sequences.

Output & Privacy Format

☒ **Anonymize Patient ID (SHA-256) & Redact Personal Info**
Hashes Patient ID and masks birth dates and approver names as REDACTED.

☒ **Simultaneously Export JSON Lines (.jsonl)**

☐ **Bypass Structure Pre-Scan**

Console Output Copy Clear

```
Ready to extract.
[INFO] Loaded preset: Prostate VMAT 786y (Standard)
[INFO] Matched 6 plans matching clinical criteria across 5 patients.
[INFO] Structure Mapping verified (PTV, Rectum, Bladder, FemoralHeads).
[INFO] Ready for execution.
```

▶ Run Extraction Cancel Idle / Ready (6 plans selected for extraction) 100%

1. Open Tab 4: 4- ⚙️ Extraction & Analysis Options.

2. Export File Destination:

- **Browse...:** Select destination folder and CSV file name (Default: `DataMiningOutput.<timestamp>.csv`).
- **Open Folder:** Open output directory in Windows File Explorer.

3. Card 1: Plan Metadata & Beam Parameters:

- Planning Approver, Approval Date, Calculation Model, Normalization Mode, Clinical Protocol, Optimization Objectives.
- Beam MU, Machine / Energy / Technique / PlanType, Calculation Logs (newlines sanitized).

4. Card 2: Complexity Analysis, Privacy & Output Formats:

- **Plan Complexity Analysis:** Computes Modulation Complexity Score (MCS), Edge Metric, Leaf Travel Length, and Arc Length.
- **Anonymize Patient ID & Redact Personal Info:** Hashes Patient ID via SHA-256 and masks birth dates and approver names as `REDACTED`.
- **Simultaneously Export JSON Lines (.jsonl):** Exports hierarchical `.jsonl` file alongside CSV for ML / AI pipelines.
- **Bypass Structure Pre-Scan:** Bypasses the structure pre-scan confirmation dialog.

Step 5: Execute Data Mining & Monitor Progress

1. Click the large ocean-blue "► **Run Extraction**" button in the bottom control bar.
 - **If plans were selected in Tab 1:** Only plans with `Extract = true` are extracted.
 - **If search was bypassed:** All matching plans across the database are directly processed.
2. The operational status badge updates to `EXTRACTING`, and the progress bar reflects real-time progress.
3. **Dark Terminal Console:**
 - Real-time logging of patient opening, DVH sampling, and memory collection using `ConsoLas` font.
 - Click "📄 **Copy**" to copy logs to clipboard or "🗑️ **Clear**" to clear display.
4. **Instant Safe Cancellation (■ Cancel):**
 - Click the red "■ **Cancel**" button to abort extraction safely.
 - Cooperative cancellation stops execution within milliseconds without file corruption. All plans extracted prior to cancellation remain safely saved on disk.
5. Upon completion, a summary dialog appears, and output files are ready for analysis.

4. Output Data Specifications & Comprehensive Column Reference

EclipseDataMiner exports a **normalized flat CSV (1 row = 1 plan, UTF-8 with BOM)** and a **hierarchical JSON Lines (`.jsonl`)** format designed for seamless ingestion into Python, R, and machine learning pipelines.

4.1 Normalized Flat CSV Output - Full Column Reference

Each row in the exported CSV consists of the following logical column groups in order:

1. **Core Plan Information Columns** (11 standard columns, always exported)
2. **Optional Plan Metadata & Beam Parameter Columns** (up to 10 columns enabled via Tab 4 checkboxes)
3. **Automatic Baseline Structure Statistics** (4 standard columns per mapped structure alias)
4. **Dynamic Dose Quality Parameters (DQP) Columns** (custom DVH metrics configured in Tab 3)

1. Core Plan Information Columns (Core Columns: Always Exported)

Column Header	Data Type / Unit	Detailed Description & Clinical Specification	Example / Default Value
Patient ID	String (string)	Patient identification number. When "Anonymize Patient ID" is enabled in Tab 4, replaced with an irreversible salted SHA-256 hash (64 hex characters).	12345678 or e3b0c44298fc1c149afb4...
Course ID	String (string)	Eclipse course identifier (<code>Course.Id</code>).	C1 , Course1
Date of birth	Date (yyyy-MM-dd)	Patient date of birth. Masked as REDACTED when anonymization is enabled. Defaults to N/A if unrecorded.	1955-04-12 or REDACTED
Plan ID	String (string)	Treatment plan identifier (<code>PlanSetup.Id</code> or <code>PlanSum.Id</code>).	VMAT_Prostate , PlanSum1
Target volume	String (string)	Target structure ID linked to the plan (<code>TargetVolumeId</code>). Defaults to N/A if unassigned.	PTV_78Gy , PTV
DosePerFraction[Gy]	Float [Gy]	Prescribed dose per fraction. Automatically unified to Gray [Gy] rounded to 2 decimal places. N/A if uncalculated.	2.00
NumberOfFractions	Integer (int)	Number of treatment fractions. Defaults to N/A if uncalculated.	39 , 35
TotalDose[Gy]	Float [Gy]	Total prescribed dose (<code>DosePerFraction * NumberOfFractions</code>). Unified to Gray [Gy] (2 decimal places). N/A if uncalculated.	78.00 , 70.00
NumberOfBeams	Integer (int)	Total number of treatment beams (excluding setup beams). For plan sums (<code>PlanSum</code>), outputs 0 .	2 , 4
ApprovalStatus	String (string)	Plan approval status in Eclipse: TreatmentApproved , PlanApproved , Unapproved , or PlanSum .	TreatmentApproved
IsPlanSum	Boolean (bool)	Boolean flag identifying whether the record is a plan sum (<code>True</code>) or a single plan setup (<code>False</code>).	True or False

2. Optional Plan Metadata & Beam Parameter Columns (Optional Columns)

Appended dynamically based on checkboxes in Tab 4 ("Plan Metadata", "Beam Parameters & Calculation Logs", "Plan Complexity Analysis"):

Column Header	Enabling Option	Detailed Description & Clinical Specification	Example / Format
PlanningApprover	Planning Approver	Eclipse username of staff who approved the plan. Masked as REDACTED when anonymized; N/A if unapproved.	physicist01 or REDACTED
PlanningApprovalDate	Planning Approval Date	Plan approval timestamp (yyyy-MM-dd HH:mm:ss). Defaults to N/A if unapproved.	2026-06-15 14:30:00
MU	Beam MU	Monitor Units (MU) for each beam concatenated with semicolons (;). Format: BeamID:MU . Outputs N/A for PlanSum.	B1:145.2;B2:138.6
Machine/Energy/Tech/PlanType	Machine / Energy / Tech / PlanType	Hardware and delivery technique specification per beam concatenated with semicolons. Format: BeamID(Machine/Energy/Tech/PlanType) .	B1(TrueBeam/6X/ARC/VMAT);B2(TrueBeam/6X/ARC/VMAT)
CalculationModel	Calculation Model	Dose calculation algorithm models for photon and electron models delimited by slash (/).	AAA_16.1.0/N/A or AcurosXB_16.1.0/N/A
CalculationLog	Calculation Logs	Execution log of dose calculation algorithm. All internal newlines (\r\n , \n) are sanitized into single spaces to preserve CSV integrity. N/A if absent.	Calculation completed in 12.4s; Grid: 2.5mm ...
PlanNormalizationMethod	Normalization Mode	Description string of dose normalization mode (e.g., target volume mean, specific point).	100.0% in target PTV
ClinicalProtocol	Clinical Protocol	Associated clinical protocol or clinical trial template name.	JC061408_Lung , RT060534
OptimizationObjectives	Optimization Objectives	Semicolon-delimited list of all optimization objectives. Format: Structure:ObjectType:Dose/Volume:Priority . Outputs N/A if absent (e.g. 3D-CRT).	PTV:Upper:70.0Gy:100;Rectum:Upper:50.0Gy:80
PlanComplexity	Plan Complexity	Literature-validated geometric beam delivery complexity metrics aggregated across all treatment beams: MCS : Modulation Complexity Score (Masi 2013 [0–1], closer to 1 is simpler) EdgeMetric : Perimeter-to-area ratio (Younge 2012 [mm⁻¹]) LeafTravel : Total leaf travel distance [mm]	MCS=0.342;EdgeMetric=0.087;LeafTravel=1420.5;ArcLength=358.0;AAV=0.512;LSV=0.668

Column Header	Enabling Option	Detailed Description & Clinical Specification	Example / Format
		- ArcLength : Total gantry rotation angle [deg] - AAV : Area Aperture Variation - LSV : Leaf Sequence Variation	

3. Automatic Baseline Structure Statistics (Automatic Baseline Structure Statistics)

For every structure mapped in Tab 2 (`IsSelected = true`), **these four fundamental DVH statistics are automatically exported without requiring manual DQP metric entry:**

Column Header	Unit	Detailed Description & Clinical Specification	Example
<Alias>-Volume[cc]	cc (cm ³)	Geometric structure volume rounded to 2 decimal places. Outputs N/A if structure is absent in the plan.	42.50
<Alias>-Max dose[Gy]	Gray [Gy]	Maximum dose within the structure. Automatically unified from cGy to Gy rounded to 2 decimal places.	81.25
<Alias>-Mean dose[Gy]	Gray [Gy]	Mean dose within the structure in Gray [Gy] (2 decimal places).	78.40
<Alias>-Min dose[Gy]	Gray [Gy]	Minimum dose within the structure in Gray [Gy] (2 decimal places).	71.10

Example: If **PTV** and **Rectum** are mapped as Target Aliases, 8 baseline columns are automatically generated: **PTV-VoLume[cc]**, **PTV-Max dose[Gy]**, **PTV-Mean dose[Gy]**, **PTV-Min dose[Gy]**, **Rectum-VoLume[cc]**, **Rectum-Max dose[Gy]**, **Rectum-Mean dose[Gy]**, **Rectum-Min dose[Gy]**.

4. Dynamic Dose Quality Parameters (DQP) Columns (Dynamic DQP Columns)

Each dose-volume parameter registered in Tab 3 is sampled in real-time from the 3D dose grid (DVH) and exported using the configured output unit:

Column Header Pattern	Metric Type (DQP Type)	Calculation Logic & Details	Example
<Alias>-D<Value>%[Gy]	Dose at Volume (D)	Dose [Gy] received by specified relative volume (%).	PTV-D95%[Gy] -> 74.50
<Alias>-D<Value>cc[Gy]	Dose at Volume (D)	Dose [Gy] received by specified absolute volume (cc). Standard for small-volume hotspot evaluation.	SpinalCord-D0.1cc[Gy] -> 42.10
<Alias>-D<Value>%[%]	Dose at Volume (D)	Relative dose [%] received by specified volume (normalized against Total Prescription Dose).	PTV-D98%[%] -> 96.50
<Alias>-V<Value>Gy[%]	Volume at Dose (V)	Volume percentage [%] receiving specified dose [Gy]. Standard for OAR tolerance evaluation.	Rectum-V70Gy[%] -> 12.30
<Alias>-V<Value>%[%]	Volume at Dose (V)	Volume percentage [%] receiving specified percentage of prescription dose.	Bladder-V70%[%] -> 25.40
<Alias>-V<Value>Gy[cc]	Volume at Dose (V)	Absolute volume [cc] receiving specified dose [Gy].	Rectum-V50Gy[cc] -> 15.20
<Alias>-DC<Value>%[Gy]	Dose Complement (DC)	Dose [Gy] received by complement volume.	PTV-DC95%[Gy] -> 72.80
<Alias>-CV<Value>Gy[%]	Complement Volume (CV)	Complement volume percentage [%] kept below specified dose.	Body-CV10Gy[%] -> 85.20

NOTE**Relative Dose Calculation Behavior in PlanSum (Composite Plans)**

- **InputUnit = Relative (%)** (e.g., V70%[Gy] , V50%[%]):

Because PlanSum lacks a single prescription dose (TotalDose), the calculation falls back to using the **structure's maximum dose (Max dose)** as the 100% reference dose (e.g., V50% evaluates volume receiving 50% of the structure's maximum dose).

- **OutputUnit = Relative (%)** (e.g., D95%[%]):

Outputs N/A safely if composite normalization reference dose is undefined in Eclipse.

- **Best Practice:**

When extracting data from cohorts containing PlanSum , configuring metrics in **Absolute Dose (Gy)** (e.g., V50Gy[%] , D95%[Gy] , D0.1cc[Gy]) is strongly recommended for clinical consistency.

4.2 Special Characters, Sanitization, and Missing Value Specifications

- **Missing Values (N/A):** If a structure does not exist in a plan, dose calculation is missing, or a parameter is inapplicable (such as beam parameters for a PlanSum), the field outputs explicit N/A rather than an

empty string.

- **Newline Sanitization:** Newline characters (`\r\n` , `\n`) in multiline fields like calculation logs are replaced with single spaces to protect CSV structure.
- **Comma & Quote Escaping:** Any field containing commas (`,`) or double quotes (`"`) is enclosed in double quotes `" "` , and internal quotes are escaped as `" "` .
- **Character Encoding:** Encoded as **UTF-8 with BOM** (Byte Order Mark) to ensure native opening in Microsoft Excel without character encoding corruption.

4.3 Python (Pandas) / JSONL Data Analysis Examples

Loading CSV in Python (Pandas)

```
import pandas as pd

# Load extracted CSV (UTF-8 with BOM parsed seamlessly)
df = pd.read_csv("DataMiningOutput.20260928120000.csv")
print(f"Loaded {len(df)} plans.")

# Missing values (N/A) are parsed as NaN automatically
print(df[["Patient ID", "Plan ID", "TotalDose[Gy]", "PTV-D95%[Gy]"]].head())

# Descriptive statistics of dosimetric endpoints
print(df["PTV-D95%[Gy]"].describe())
```

Loading JSON Lines (`.jsonl`) in Python

```
import json

plans = []
with open("DataMiningOutput.20260928120000.jsonl", "r", encoding="utf-8") as f:
    for line in f:
        plans.append(json.loads(line))

print(f"Loaded {len(plans)} plans.")
# Directly access hierarchical beam parameters and optimization objectives:
first_plan_beams = plans[0]["Beams"]
```

5. Troubleshooting & FAQ

Symptom	Cause	Solution
Startup Error	Missing Eclipse ESAPI runtime on workstation	Ensure the executable is run on an authorized Eclipse clinical workstation or TBox.
Slow Pre-Scan	Unfiltered pre-scan on large database	Use Patient ID, date range, or approval filters to narrow target plans.
Structure Not Exported	Case or underscore naming mismatch	Execute Pre-Scan in Tab 2 to verify raw Structure IDs and map to Target Alias.
Output File Locked	CSV file is open in Microsoft Excel	Close Excel or select a new file name using Browse
PlanSum DQP Relative Dose is N/A or Unexpected	PlanSum lacks Total Prescription Dose (TotalDose)	Define metrics using absolute doses (Gy , e.g., D95%[Gy] , V50Gy[%]) instead of relative percentages.
Abort Extraction	Stop long-running query safely	Click " ■ Cancel ". Execution safely halts within 1 second and saves all completed records.

Chapter 3

System Architecture Specification

Layered architecture, STA worker thread model, streaming export pipeline, and memory management

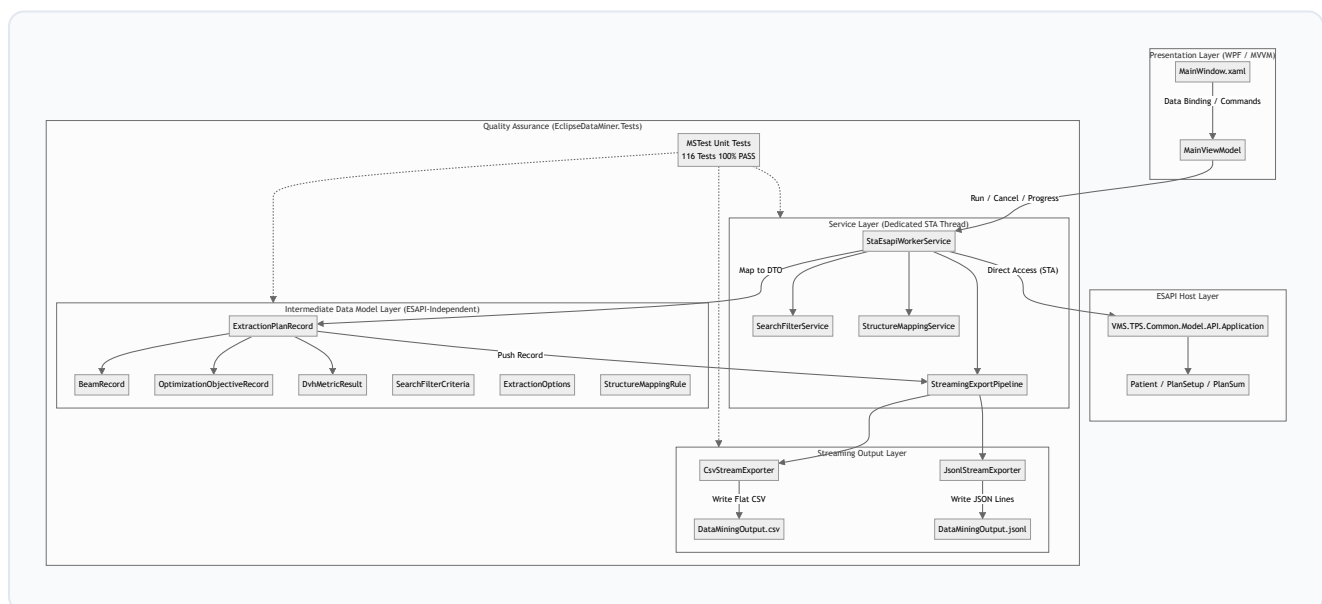
1. High-Level Architectural Overview

EclipseDataMiner is a standalone WPF application built on the Varian Eclipse Scripting API (ESAPI) to safely and rapidly extract dosimetric, geometric, and metadata parameters from several thousand to 10,000+ radiation therapy plans.

The system is built on **4 core design principles**:

- MVVM Pattern & Loose DTO Coupling:** Clean separation between presentation (UI), business logic, ESAPI dependencies, and streaming export layers.
- Dedicated STA Worker Thread Isolation:** Full compliance with ESAPI Single-Threaded Apartment (STA) constraints; all queries execute on an isolated background thread without blocking the UI.
- Streaming Export Pipeline:** Prevents memory bloat by immediately flushing extracted records to disk on a per-plan / per-patient basis.
- Self-Contained Single Executable:** NuGet dependencies are embedded into `EclipseDataMiner.exe` via `Costura.Fody` for seamless zero-install deployment on Eclipse workstations.

2. Layer Architecture Diagram



3. Threading Model and Memory Management

3.1 Dedicated STA Worker Thread

- ESAPI objects must be created and queried on a Single-Threaded Apartment (STA) thread.
- To eliminate UI freezing, a dedicated background thread is spawned using `Thread.SetApartmentState(ApartmentState.STA)`.
- `Application.CreateApplication()` is instantiated inside this worker thread and cleanly disposed upon extraction completion or cancellation.
- Communication with the UI occurs exclusively via thread-safe `IProgress<ExtractionProgressInfo>` messaging.

3.2 Strict Memory Management (LOH & Unmanaged Leak Prevention)

1. Deterministic Patient Disposal:

- Only a single patient is opened in memory at any given time.
- `Application.ClosePatient()` is guaranteed to be invoked within a `try ... finally` block, preventing patient locks and lingering resources even if an exception occurs.

2. Periodic Garbage Collection:

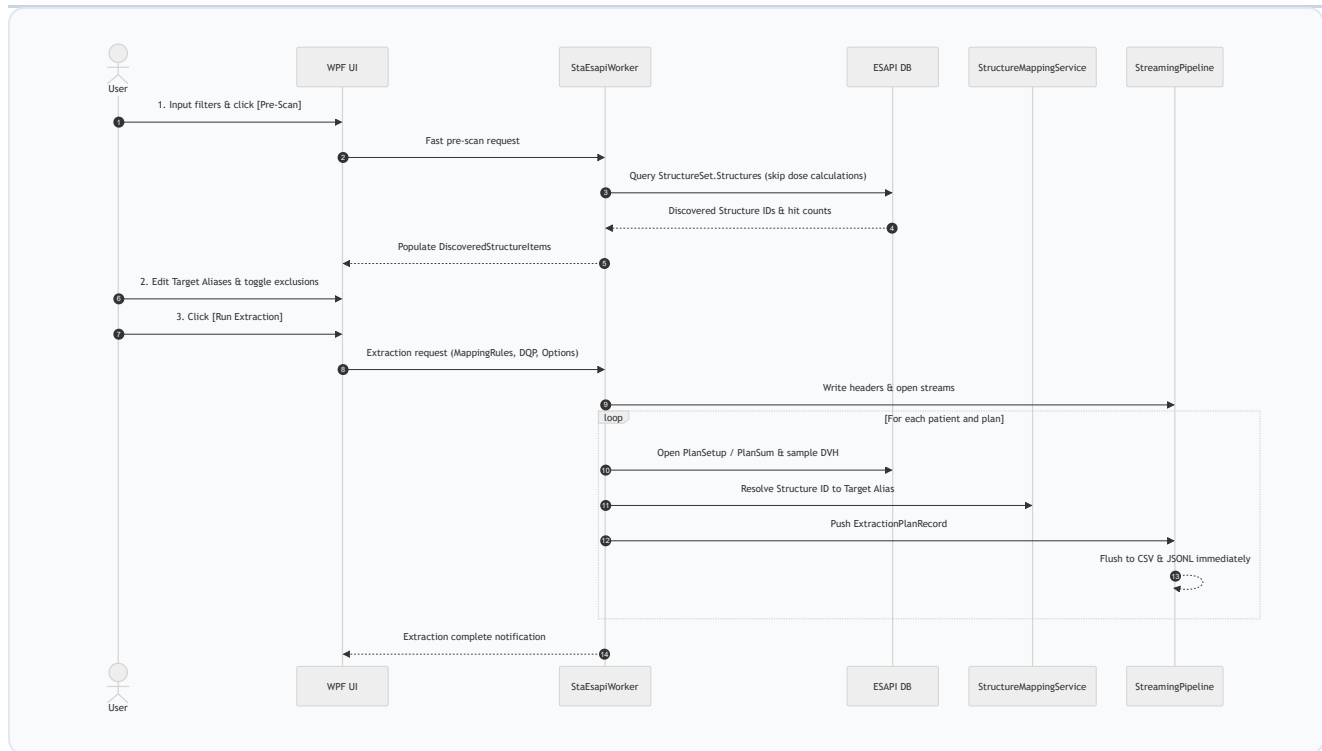
- Explicit calls to `System.GC.Collect()` and `System.GC.WaitForPendingFinalizers()` occur every 200 patients, purging unmanaged memory caches and ESAPI COM wrappers.

3. Streaming Disk Export:

- Extracted plan records are immediately flushed to disk via `StreamWriter.WriteLine()` rather than accumulated in memory. Working set memory remains flat (under 200 MB) even during 10,000+ plan queries.

4. Structure Pre-Scan & Nomenclature Mapping Architecture

To address the primary challenge of clinical data mining—structure nomenclature variations (e.g., `PTV_60`, `ptv60`, `PTV-60Gy`)—EclipseDataMiner implements a two-stage pipeline:



5. Output Format Specifications

5.1 Normalized Flat CSV

- **Encoding:** UTF-8 (BOM supported)
- **Delimiter:** Comma (,)
- **Granularity:** 1 row per plan (flat tabular format).
- **One-to-Many Aggregation:** Multi-beam records and optimization objectives are joined by semicolons (;) into single cells.
- **Sanitization:** Newlines (\r\n , \n) are replaced with spaces, and fields containing commas or quotes are escaped with double quotes (" ").
- **Missing Values:** Uncalculated or non-applicable fields output N/A .

5.2 JSON Lines (.jsonl)

- **Granularity:** 1 JSON object per line.
- **Hierarchical Fidelity:** Preserves full object structures (Beams , OptimizationObjectives , DvhMetrics) without flattening.
- **Target Applications:** Tailored for Python data science (Pandas, Polars), PyTorch, and machine learning pipelines.

5.3 Patient De-Identification & Privacy

- When AnonymizeOutput is enabled:
 - Patient ID : Replaced with salted SHA-256 hash (64-character hexadecimal string).
 - DateOfBirth , PlanningApprover : Replaced with REDACTED .

6. Packaging and Binary Compatibility

- **Target Framework:** .NET Framework 4.6.1 (Fully compatible with Eclipse v15.6 and v16.1)
- **Platform Architecture:** x64 (AMD64)
- **Costura.Fody Embedding:**
 - Bundles dependencies (`CommunityToolkit.Mvvm.dll` , `System.Text.Json.dll` , etc.) directly inside `EclipseDataMiner.exe` .
 - Excludes `VMS.TPS.Common.Model.API.dll` and `VMS.TPS.Common.Model.Types.dll` via `ExcludeAssemblies` to dynamically link against workstation GAC / official installation assemblies.

Chapter 4

Detailed Design Specification

Input/output data specifications (CSV/JSONL), plan complexity metrics (MCS, Edge Metric, LT, AL), and de-identification

1. System Overview

This software is a standalone WPF application designed to rapidly and safely extract DVH metrics and metadata from several thousand to 10,000+ radiation therapy plans (PlanSetup / PlanSum) utilizing the Varian Eclipse Scripting API (ESAPI). Extracted records are streamed into normalized CSV and machine-learning-ready JSON Lines (JSONL) formats to minimize downstream data cleansing efforts.

2. Architectural Design

- **Target Platform:** .NET Framework 4.6.1 (Eclipse v16.1 compatible) / x64 Target
 - Project format: SDK-style csproj / `<LangVersion>10.0</LangVersion>`
- **UI Framework:** Windows Presentation Foundation (WPF)
- **Design Pattern:** MVVM Pattern (`CommunityToolkit.Mvvm`)
- **Project Structure:**
 - Single main project (`EclipseDataMiner`) + unit test project (`EclipseDataMiner.Tests`)
 - Clean layering: Models (DTO), ViewModels, Views, Services, Helpers
- **Decoupled Intermediate Data Model (DTO) Architecture:**
 - ESAPI objects are immediately mapped into independent DTO models (such as `ExtractionPlanRecord`).
 - CSV/JSONL serialization, sanitization, and aggregation logic operate purely on DTOs, enabling 100% offline unit testing without ESAPI connections.
- **Asynchronous Threading Model:**
 - A dedicated Single-Threaded Apartment (STA) worker thread runs all background operations to keep the UI completely responsive.
 - In compliance with ESAPI STA constraints, plan iteration executes strictly sequentially. Concurrent operations such as `Parallel.ForEach` are prohibited.
- **Streaming Data Export Pipeline:**
 - Records are flushed to disk per patient/plan, keeping memory usage flat regardless of total query size.
- **Packaging:**
 - `Costura.Fody` bundles external NuGet dependencies into a single standalone binary (`EclipseDataMiner.exe`), excluding ESAPI DLLs.

3. Functional Requirements

3.1. Search & Filtering Engine

- **Hierarchical AND / OR Search:**
 - **In-field OR Search:** Comma-separated entries evaluate with OR logic (e.g., PlanID = VMAT, IMRT).
 - **Global Logic Switch:** Toggles between AND (all criteria must be satisfied) and OR (any criterion matches).
- **Target Attributes:** Patient ID, Course ID, Plan ID, Dose/Fr, Fractions, Total Dose, Approval Status.
- **Composite PlanSum Support:**
 - Optional Include PlanSum checkbox (default: off).
 - Automatically populates non-applicable fields (e.g., beam logs and optimization objectives) with N/A .

3.2. Structure Pre-Scan & Nomenclature Mapping (Two-Pane Layout with Real-Time Preview)

- **Two-Pane Architecture:**
 - **Left Pane (Mapping Rules):** Persistent rule dictionary (Pattern , MatchMode , TargetAlias , IsSelected). Pre-scanning does not overwrite loaded rules. [▲ Up] / [▼ Down] buttons control evaluation precedence.
 - **Right Pane (Discovered Structures & Preview):** Collects all raw Structure.Id values and hit counts across matching plans, rendering a real-time preview of resolved aliases and status.
- **Rule Conflict Resolution (Specificity Priority + Top-First):**
 1. **Exact Match** (Exact) evaluates first.
 2. **Contains Match** (Contains) evaluates second.
 3. **Regex Match** (Regex) evaluates third.
 4. Within the same match mode, the **topmost rule** in the list takes priority (reordered with Up/Down buttons).
 5. Duplicate rules with identical Pattern and MatchMode are automatically prevented.
- **Behavior with Multiple Raw Structures Mapped to the Same Target Alias:**
 - **Intentional Variation Consolidation:** Consolidates raw structures across patients into single CSV columns (e.g., PTV-Volume[cc] , PTV-D95%[Gy]).
 - **Collisions Within the Same Plan:** If multiple raw structures within a single plan match the same alias, the first matching structure is output to CSV, while JSON Lines preserves all structures hierarchically.
 - **Visual Annotation:** Discovered structures preview displays [Merged: N] when multiple raw structures map to the same alias.
- **Persistence:** Rules persist to JSON files.
- **One-Click Rule Creation:** Double-clicking a discovered structure row or clicking + Add to Rules adds a new rule and refreshes the preview.
- **Bypass Option:** Supports bypassing pre-scan confirmation for routine workflows.

3.3. Extracted Parameters

- **Automatic Baseline DVH Statistics:** `Volume[cc]` , `Max dose[Gy]` , `Mean dose[Gy]` , and `Min dose[Gy]` are exported automatically for all mapped structures without manual DQP configuration.
- **Dynamic DQP Metrics:** 5-column format (`structureName` , `DQPtype` , `DQPvalue` , `InputUnit` , `OutputUnit`) supporting `Dose` (D), `Volume` (V), `DoseComplement` (DC), and `ComplementVolume` (CV).
- **Prescription Dose Normalization:** Relative dose parameters (e.g., V70%) normalize against the plan's Total Prescription Dose (`PlanSetup.TotalDose`).
- **Delivery Parameters:** Beam MU, Machine, Energy, Technique, PlanType (multi-beam records joined by semicolons).
- **Plan Metadata:** Calculation Model, Normalization Mode, Clinical Protocol, Optimization Objectives.
- **Calculation Logs:** Exported with newlines replaced by spaces.
- **Plan Complexity Analysis:** Computes literature-validated complexity indices (MCS / MCSv [Masi 2013, McNiven 2010], Edge Metric [Younge 2012], Leaf Travel Length, Arc Length).
- **De-Identification:** Replaces Patient ID with salted SHA-256 hash and masks birth dates and approver names as `REDACTED` .

4. Output Data Specifications

- **Unit Normalization (Gy):** Converts `cGy` to `Gy` automatically. Column headers explicitly specify units (e.g., `[Gy]`).
- **CSV Format (1 plan = 1 row normalized flat format):**
 - **1. Core Plan Information Columns (11 columns, always exported):**
 - `Patient ID` , `Course ID` , `Date of birth` , `Plan ID` , `Target volume` , `DosePerFraction[Gy]` , `NumberOfFractions` , `TotalDose[Gy]` , `NumberOfBeams` , `ApprovalStatus` , `IsPlanSum`
 - **2. Optional Plan Metadata & Beam Parameter Columns (Up to 10 columns, enabled via Tab 4):**
 - `PlanningApprover` , `PlanningApprovalDate` , `MU` , `Machine/Energy/Tech/PlanType` , `CalculationModel` , `CalculationLog` , `PlanNormalizationMethod` , `ClinicalProtocol` , `OptimizationObjectives` , `PlanComplexity` (MCS, EdgeMetric, LeafTravel, ArcLength, AAV, LSV)
 - **3. Automatic Baseline Structure Statistics (4 columns per mapped structure alias):**
 - `<Alias>Volume[cc]` , `<Alias>Max dose[Gy]` , `<Alias>Mean dose[Gy]` , `<Alias>Min dose[Gy]`
 - **4. Dynamic Dose Quality Parameters (DQP) Columns (Configured in Tab 3):**
 - `<Alias>D<Val>[%Gy]` , `<Alias>D<Val>cc[Gy]` , `<Alias>V<Val>Gy[%]` , `<Alias>DC<Val>[%Gy]` , `<Alias>CV<Val>Gy[%]` , etc.
 - Multi-value fields (e.g., beam parameters, optimization objectives) are separated by semicolons (`;`).
 - Strings containing commas or newlines (e.g., calculation logs) have newlines converted to spaces, quotes escaped as `" "` , and are wrapped in double quotes.
 - Missing or non-applicable values output as `N/A` .
 - Encoding: UTF-8 with BOM for native opening in Microsoft Excel.
- **JSONL Format:**

- 1 JSON object per line preserving hierarchical structure (`Beams` , `OptimizationObjectives`) for ML/AI workflows.

5. ESAPI Implementation & Resource Safety

- **Deterministic Patient Disposal:** At most 1 patient in memory; calls `Application.ClosePatient()` in a `finally` block.
- **Periodic GC:** Invokes `System.GC.Collect()` every 200 patients.
- **Null Safety:** Comprehensive null checks for uncalculated doses, missing structure sets, and 3D-CRT plans without optimization.

6. UI & Design System Specifications

- **Design System:** Slate & Ocean Cyan Enterprise Theme.
- **Color Tokens:**
 - Primary: `#0284C7` (Sky-600)
 - Dark Header / Console: `#0F172A` (Slate-900)
 - Surface: `#FFFFFF`
 - Danger: `#EF4444` (Red-500)
 - Fonts: `Segoe UI` (UI), `Consolas` (Console & code)
- **Components:** Single-row action bar, 4 numbered tabs, dark terminal console, and overlaid progress bar.

7. Quality Assurance (QA) & Test Specifications

The automated testing pipeline (`test.bat`) executes MSBuild x64 Release builds and MSTest suites (116 tests, 100% pass) to ensure calculation accuracy and stability.

Chapter 5

Clinical Commissioning Guide

Acceptance testing protocols, geometric & dosimetric concordance verification, and clinical sign-off form

This document establishes the **acceptance criteria, verification checklists, and quality assurance procedures** required before deploying **EclipseDataMiner (v3.0 / compatible with Eclipse v15.6 & v16.1)** for clinical research and quality management.

NOTE

All operations performed by EclipseDataMiner are strictly **Read-Only**. It never alters or saves patient or plan data (no `SaveModifications` calls). However, because it queries large volumes of treatment plans, clinical acceptance testing must be performed on a representative dataset in a test environment or TBox and approved by the Chief Medical Physicist before clinical adoption.

1. Commissioning Principles & Standards

Commissioning of EclipseDataMiner complies with established medical physics and radiation oncology safety standards:

- **AAPM TG-275:** Quality control and safety procedures for radiation oncology
- **AAPM TG-263:** Standardized nomenclature for structures and dose-volume parameters
- **ESTRO Booklet No. 9:** Guidelines for target volumes and organs at risk evaluation
- **AAPM TG-101 / ICRU Report 83:** Prescribing, recording, and reporting stereotactic and high-precision radiotherapy

Core Guidelines

1. **Strict Concordance with Ground Truth:** Values extracted by EclipseDataMiner (Volume, Max dose, Mean dose, Min dose, D95%, V20Gy, etc.) must match the Eclipse DVH Statistics window within strict tolerance (≤ 0.01 Gy / 0.01 cc / 0.01%).
2. **Prescription-Based Relative Dose Normalization:** Relative dose parameters (e.g., V70%) must be normalized strictly against the plan's Total Prescription Dose (`TotalDose`).
3. **Non-Blocking Execution & Memory Safety:** Continuous scanning of 1,000+ plans must not freeze the UI or leak memory (Working Set must remain flat).
4. **Complete Privacy Protection:** When anonymization is enabled, raw Patient IDs, birth dates, and approver names must be completely excluded from CSV and JSONL exports.

2. Test Environment & Prerequisites

Item	Requirement / Test Benchmark
TPS Version	Varian Eclipse v15.6 / v16.1
OS / Hardware	Windows 10 / 11 64-bit, 16 GB+ RAM
Runtime	.NET Framework 4.6.1 or later
User Rights	ESAPI Scripting read permissions
Benchmark Dataset	① Representative clinical cases (Prostate VMAT, Head & Neck IMRT, Lung SBRT) ② Cases containing composite plan sums (<code>PlanSum</code>) ③ Edge cases (uncalculated doses, unapproved plans, 3D-CRT plans without optimization)

3. Verification Checklists by Category

3.1 Search & Filter Criteria Validation

- **[] Comma-Separated OR Search:** Querying multiple IDs (e.g., `ID001, ID002`) retrieves all matching cases without omission.
- **[] Global Logic Switch (AND / OR):**
 - `AND` mode: Only plans meeting all criteria (dose, fractions, status, etc.) are matched.
 - `OR` mode: Plans meeting any individual criterion are matched.
- **[] Approval Status Filter:** Selection of `Unapproved` , `Plan approved` , and `TRT approved` filters plans correctly.
- **[] PlanSum Handling:**
 - `Include PlanSum` unchecked: Only `PlanSetup` instances are queried.
 - `Include PlanSum` checked: `PlanSum` instances are included; non-applicable parameters (e.g., beam logs) output `N/A` cleanly.

3.2 Structure Pre-Scan & Nomenclature Mapping Validation

- **[] High-Speed Pre-Scan:** Discovers all raw `Structure.Id` values within selected plans and counts occurrences accurately.
- **[] Exact Match:** Strict matching (e.g., `PTV-PROST`) correctly maps to the designated Target Alias.
- **[] Contains Match:** Substring matching (e.g., `prost`) consolidates variations into a unified alias.
- **[] Regex Match:** Regular expression patterns (e.g., `PTV.*|ptv.*`) match case variations and suffixes into a single `PTV` column.
- **[] Exclusion:** Unchecked structures (e.g., `BODY` , `CouchInterior`) are excluded from extraction.
- **[] Rule Persistence:** `Save Rules (JSON)` / `Load Rules (JSON)` reproduces rule dictionaries exactly (verified with `Templates/StructureMapping_Prostate.json`).

3.3 Baseline DVH Statistics & Dose Normalization (Gy)

- **[] Gy Unit Normalization:** When internal ESAPI values are in `cGy`, all CSV and JSONL outputs are converted to `Gy`.
- **[] Automatic 4 Baseline Metrics:** For every mapped structure, the following 4 columns are automatically output without explicit DQP registration:
 - `<Structure>—Volume[cc]` : Matches Eclipse structure volume (≤ 0.01 cc).
 - `<Structure>—Max dose[Gy]` : Matches Eclipse DVH Max Dose (≤ 0.01 Gy).
 - `<Structure>—Mean dose[Gy]` : Matches Eclipse DVH Mean Dose (≤ 0.01 Gy).
 - `<Structure>—Min dose[Gy]` : Matches Eclipse DVH Min Dose (≤ 0.01 Gy).

3.4 Dynamic DQP Calculation Accuracy

- **[] Dose at Volume (D):**
 - `D95%[Gy]` : 95% volume dose matches Eclipse DVH.
 - `D0.1cc[Gy]` : Absolute volume 0.1 cc dose matches Eclipse DVH.
 - `D95%[%]` : Outputs relative percentage correctly.
- **[] Volume at Dose (V):**
 - `V20Gy[%]` : 20 Gy irradiated volume percentage matches Eclipse DVH.
 - `V40Gy[cc]` : Volume in cc matches Eclipse DVH.
 - `V70%[%]` : Volume receiving 70% of prescription total dose is accurately computed.
- **[] Dose Complement (DC) & Complement Volume (CV):**
 - Accurately computes dose complement and complement volume.
- **[] DQP Template:** `Templates/DQPList_Prostate.csv` loads cleanly and generates expected columns.

3.5 Delivery Parameters & Modulation Complexity Analysis

- **[] Beam MU:** Total MU per beam is formatted as `B1:145.2;B2:138.6`.
- **[] Machine / Energy / Technique:** Machine name, energy mode (6X, 10X), delivery technique (ARC, STATIC), and plan type output correctly.
- **[] Complexity Metrics (MCS / Edge Metric):** When enabled, Modulation Complexity Score and Edge Metric are computed accurately.
- **[] Calculation Log:** Algorithm logs are formatted with newlines replaced by spaces to maintain CSV integrity.

3.6 Privacy Protection & Patient De-Identification

- **[] Patient ID Hashing:** Patient IDs are converted to irreversible 64-character SHA-256 hashes. Identical IDs produce identical hashes.
- **[] PII Masking:** Birth dates and planning approver names are replaced with `REDACTED`.

3.7 Streaming Export Pipeline & JSONL Export

- **[] Instant Disk Flush:** Records are written and flushed immediately per plan.
- **[] JSON Lines Validity:** Output `.jsonl` file is parsable line-by-line using standard JSON libraries (Python `json.loads`).

4. Stress Testing, Failsafe, and Error Handling

- [] **High-Volume Continuous Scan:** Running 1,000+ plans maintains flat memory usage (<200 MB) without `OutOfMemoryException` (GC every 200 patients).
- [] **Cooperative Cancellation:** Clicking "■ **Cancel**" aborts execution within milliseconds without data corruption.
- [] **Missing Data Resilience:** Handles uncalculated plans, plans without structure sets, and 3D-CRT plans without optimization without crashes (outputs `N/A`).
- [] **File Lock Detection:** Displays clear error messages when the destination CSV file is locked in Microsoft Excel.

5. Clinical Acceptance Sign-Off Form

Item	Evaluation	Date	Evaluator	Evidence / Notes
1. Search Criteria & PlanSum	Pass / Fail	2026/ /		
2. Structure Pre-Scan & Aliases	Pass / Fail	2026/ /		
3. Baseline DVH Stats (Gy)	Pass / Fail	2026/ /		
4. Dynamic DQP Accuracy	Pass / Fail	2026/ /		
5. Delivery Parameters & MCS	Pass / Fail	2026/ /		
6. Privacy & De-Identification	Pass / Fail	2026/ /		
7. Streaming & JSONL Pipeline	Pass / Fail	2026/ /		
8. Resilience & Cancellation	Pass / Fail	2026/ /		

Overall Assessment and Clinical Approval

EclipseDataMiner v3.0 has completed clinical acceptance testing and meets all requirements above. It is approved for clinical research and quality management operations.

- **Approval Date:** __
- **Chief Medical Physicist:** __ (Signature)
- **Department Director:** __ (Signature)

Chapter 6

Troubleshooting & FAQ Guide

ESAPI permission errors, memory optimization, nomenclature mapping resolutions, and frequently asked questions

This document provides solutions and diagnostics for common startup, configuration, UI, filtering, structure mapping, DQP metric calculation, and streaming export issues encountered when using **EclipseDataMiner (v3.0 / Eclipse v15.6 & v16.1)**.

1. Startup & Environment Issues

1.1 ESAPI DLL Missing (`FileNotFoundException`)

- **Symptom:** Launching `EclipseDataMiner.exe` triggers `Could not load file or assembly 'VMS.TPS.Common.Model.API.dll'` and terminates immediately.
- **Cause:**
 - The application is running on a non-Eclipse workstation or the ESAPI DLLs are not in the standard GAC or application directory.
 - ESAPI DLLs are excluded from `Costura.Fody` embedding due to Varian licensing policies.
- **Resolution:**
 - **Run on Authorized Workstations:** Run the executable on an official Eclipse clinical workstation or TBox.
 - **Manual DLL Placement:** For development/testing environments, copy `VMS.TPS.Common.Model.API.dll` and `VMS.TPS.Common.Model.Types.dll` into the same folder as `EclipseDataMiner.exe`.

1.2 .NET Framework Version Mismatch

- **Symptom:** `.NET Framework Initialization Error` appears on launch.
- **Cause:** The workstation does not have .NET Framework 4.6.1 or later installed.
- **Resolution:** Install .NET Framework 4.6.1 or later (4.8 recommended) via Windows Features or the Microsoft website.

1.3 Scripting Permissions / Standalone Access Denied

- **Symptom:** Application fails with `Access Denied` or `User has insufficient rights`.
- **Cause:** The current Windows user account does not have ESAPI standalone read access.
- **Resolution:** Request your Eclipse System Administrator to grant ESAPI read rights to the account.

2. Search & Extraction Issues

2.1 UI Freezing or Becoming Unresponsive

- **Symptom:** Window title shows "Not Responding" during long queries.
- **Cause:** In legacy versions (v1.0), queries ran directly on the UI thread.
- **Resolution:** Update to **v3.0**. All queries run on an isolated background STA worker thread (`StaEsapiWorkerService`). The UI remains smooth and responsive throughout.

2.2 Out of Memory (`OutOfMemoryException`) on Large Datasets

- **Symptom:** Memory usage continuously escalates until the application crashes.
- **Cause:** Accumulating thousands of plan records in RAM.
- **Resolution:** v3.0 employs a streaming pipeline. Only 1 patient is held in memory at a time, records are immediately flushed to disk, and `GC.Collect()` is explicitly called every 200 patients. Working set memory stays flat (<200 MB).

2.3 Safely Aborting Extraction / Data Preservation on Cancellation

- **Symptom:** Need to stop a running query without losing progress.
- **Resolution:** Click the red "■ Cancel" button. The background loop cooperatively terminates within milliseconds. All records written prior to cancellation remain safely stored in the output CSV and JSONL files.

2.4 Extraction Aborting on Specific Patients or Corrupted Plans

- **Symptom:** A single corrupted plan crashes the entire run.
- **Resolution:** v3.0 wraps each patient in comprehensive `try-catch` blocks. If an uncalculated or corrupted plan is encountered, an error message is logged to the console, non-applicable fields are filled with `N/A` , and execution safely proceeds to the next patient.

3. DQP, Structure Mapping, and Export Issues

3.1 All DQP Columns Output `N/A` in CSV

- **Symptom:** Baseline metadata is exported, but DQP columns contain only `N/A` .
- **Cause:** The structure name specified in the DQP list does not match the resolved Target Alias from Step 2.
- **Resolution:** Verify that the Structure Name in Tab 3 exactly matches the `Target Alias` defined in Tab 2.

3.2 Structure Variations Not Consolidating

- **Symptom:** `PTV_60` and `ptv-60` output as separate columns.
- **Cause:** Rules in Tab 2 are configured with `Exact` mode instead of `Contains` or `Regex` .
- **Resolution:** Change the Match Mode to `Contains` (e.g., `PTV` with alias `PTV`) or use a regex pattern (e.g., `PTV.*`).

3.3 "The process cannot access the file because it is being used by another process"

- **Symptom:** Export fails immediately upon start.

- **Cause:** The destination CSV file is open in Microsoft Excel or another program.
- **Resolution:** Close Excel or specify a new file name using "**Browse...**" in Tab 4.

3.4 CSV Text Encoding / Japanese Garbled in Excel

- **Symptom:** Non-ASCII characters display incorrectly in Microsoft Excel.
- **Cause:** Excel expects a UTF-8 BOM when opening CSV files by double-clicking.
- **Resolution:** EclipseDataMiner exports UTF-8 with BOM by default. In Excel, you can also use **Data > From Text/CSV** and select UTF-8 encoding.

4. Frequently Asked Questions (FAQ)

- **Q: Does EclipseDataMiner modify clinical data in Eclipse?**
A: Absolutely not. EclipseDataMiner is strictly **Read-Only** and never invokes **SaveModifications()**.
- **Q: Can I run queries across multiple LINACs?**
A: Yes. Use the **Machine ID** filter under **Advanced Filters** (e.g., **TrueBeam, Clinac_iX**).
- **Q: Where are my search presets stored?**
A: Presets are saved in the **Presets/** folder located in the same directory as **EclipseDataMiner.exe**. You can easily copy and share them between workstations.

Chapter 7

Developer Guide & Coding Standards

MSBuild x64 build instructions, unit testing guidelines, threading rules, and pull request workflow

This document provides developer guidelines, build instructions, coding standards, and testing procedures for contributors and medical physicists developing **EclipseDataMiner**.

1. Development Environment Requirements

- **Operating System:** Windows 10 / 11 (64-bit)
- **IDE:** Visual Studio 2022 Community / Professional or later
 - Required Workload: **.NET desktop development**
 - .NET Framework 4.6.1 Developer Pack / Targeting Pack
- **Target Platform:** .NET Framework 4.6.1 / x64
- **Language Version:** C# 10.0 (<LangVersion>10.0</LangVersion>)
- **External ESAPI Libraries:**
 - VMS.TPS.Common.Model.API.dll (v15.6 or v16.1)
 - VMS.TPS.Common.Model.Types.dll (v15.6 or v16.1)
 - *Project files automatically probe the following fallback paths:*
 1. C:\Program Files\Varian\RTM\16.1\esapi\API
 2. Parent repository directory (..\..\)
 3. Solution parent directory (\$(SolutionDir)..\)

2. Repository Setup and Build Instructions

NuGet Package Restoration

The solution uses standard `PackageReference` format. Run MSBuild to restore packages:

```
# Restore NuGet packages
& "C:\Program Files\Microsoft Visual Studio\2022\Community\MSBuild\Current\Bin\MSBuild.exe" `
    EclipseDataMiner.sln /t:Restore /verbosity:minimal
```

Building the Solution

ESAPI is a native 64-bit library; always build for the `x64` platform:

```
# x64 Release Build
& "C:\Program Files\Microsoft Visual Studio\2022\Community\MSBuild\Current\Bin\MSBuild.exe" `
    EclipseDataMiner.sln /p:Configuration=Release /p:Platform=x64 /t:Build /verbosity:minimal
```

The resulting standalone executable is generated at `EclipseDataMiner\bin\x64\Release\EclipseDataMiner.exe` (~590 KB).

3. Automated Verification Pipeline (`test.bat`)

Always run `test.bat` before submitting changes:

```
test.bat
```

The pipeline executes 5 automated stages:

1. **NuGet Package Restoration**
2. **x64 Release MSBuild Compilation**
3. **MSTest Unit Test Execution (116/116 tests 100% pass)**
4. **Release Artifact Synchronization**
5. **Release Binary Startup Smoke Testing**

4. Architectural Design Guidelines

1. **Decoupled Intermediate DTOs:** Map ESAPI objects into independent DTOs (`ExtractionPlanRecord`). Do not pass ESAPI objects to serialization or UI layers.
2. **Dedicated STA Thread:** Keep all ESAPI invocations isolated on a Single-Threaded Apartment worker thread (`StaEsapiWorkerService`).
3. **Deterministic Memory Cleanup:** Deterministically close patients inside `try-finally` blocks and call `GC.Collect()` periodically.

5. Coding Standards

- **Language:** All code comments, XML docstrings, and test descriptions must be written in **English**.
- **Naming:** Follow standard Microsoft C# naming conventions (PascalCase for classes/methods, camelCase for local variables).
- **Null Safety:** Perform strict null checks on all nullable ESAPI properties.
- **Unit Testing:** Adhere to the Arrange-Act-Assert (AAA) pattern.

6. Commit Message Conventions (Conventional Commits)

Format commit messages according to the Conventional Commits specification:

```
<type>(<scope>): <short description>
```

- `feat` : New clinical feature or extraction metric
- `fix` : Bug fix or logic correction
- `docs` : Documentation updates
- `test` : Adding or modifying unit tests
- `refactor` : Code improvements without behavior changes

7. Pull Request (PR) Checklist

- ☐ `test.bat` passes 100% (all 116 unit tests pass).
- ☐ No compilation warnings on x64 Release.
- ☐ Code comments are written in English.
- ☐ Documentation (Markdown) is updated in both English and Japanese.
- ☐ No institutional passwords, local paths, or patient identifiers are committed.

Chapter 8

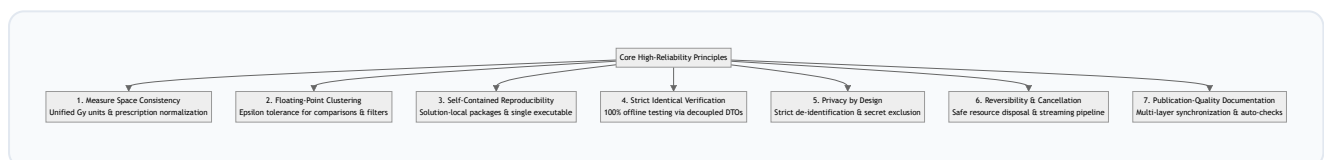
Standard Development Lifecycle Protocol (SDLP)

7 Core Quality Principles (measure space consistency, epsilon clustering, etc.), 4-layer DoD gates, and release policies

This document formalizes the standard software development lifecycle protocol and engineering quality principles designed to achieve both **maximum development efficiency** and **absolute mathematical accuracy and robustness** for large-scale Varian Eclipse ESAPI data mining applications.

1. Core Architectural Philosophy & 7 Quality Principles

To eliminate critical inconsistencies, mathematical breakdowns, resource leaks, data confidentiality breaches, and engineering regressions, all development adheres strictly to the **7 Core Principles**:



1.1 Measure Space Consistency Principle

- **Unified Gy Units:** All internal dose models, CSV columns, and JSONL records standardize strictly on `Gy` (using `DoseNormalizationHelper.ToGy()`).
- **Prescription Total Dose Normalization:** Relative dose parameters (e.g., V70%) normalize against the plan's prescription Total Dose (`PlanSetup.TotalDose`).

1.2 Floating-Point Clustering Principle

- **Epsilon Clustered Comparison:** Prohibits raw `=` comparisons on floating-point quantities (Gy, cc, MU). All boundary queries enforce an epsilon tolerance ($\epsilon = 10^{-4}$).

1.3 Self-Contained Reproducibility Principle

- **Local Package Management:** NuGets are restored into the solution-relative `packages/` directory (`nuget.config`).
- **Single Standalone Executable:** Generates an all-in-one binary via `Costura.Fody` (excluding ESAPI DLLs).

1.4 Strict Identical Verification Principle

- **DTO-Based Decoupling:** Maps ESAPI structures into plain C# models (`ExtractionPlanRecord`), validating CSV/JSONL serialization and filtering logic offline via 116 MSTest unit tests (AAA pattern).

1.5 Privacy by Design Principle

- **De-Identification Pipeline:** Irreversible salted SHA-256 hashing for Patient IDs, and **REDACTED** masking for birth dates and staff names.
- **Secret Scrubbing:** Zero internal IP addresses, local usernames, or UNC paths committed to source code or documentation.

1.6 Reversibility & Cancellation Principle

- **Isolated STA Worker Thread:** Background worker thread prevents UI blocking.
- **Resource Safety:** Deterministic `ClosePatient()` in `finally` blocks, periodic `GC.Collect()` every 200 patients, and instantaneous cooperative cancellation.

1.7 Publication-Quality Documentation Principle

- **Synchronized Documentation:** Full synchronization across source code, settings, and documentation in both English and Japanese.

2. 4-Layer Definition of Done (DoD) Quality Gates

```
[Layer 1: Build Integrity Gate]
- MSBuild (Release|x64) 0 warnings, 0 errors
- Costura.Fody single EXE verification (ESAPI excluded)
↓
[Layer 2: Automated Unit Testing Gate]
- MSTest 100% (116/116) PASS
- Complex geometry & DVH benchmark validation
↓
[Layer 3: Delivery Packaging Gate]
- Release artifacts synchronized to release/
- Startup smoke testing verified
↓
[Layer 4: Documentation Synchronization Gate]
- README, MANUAL, and technical specifications updated in both English and Japanese
```

Appendix

Release History (Changelog)

v3.0.0 / v2.4.0 / v2.3.0 release notes, key milestones, and historical changelog

All notable changes to this project will be documented in this file. The format is based on [Keep a Changelog](#), and this project adheres to [Semantic Versioning](#).

[Unreleased]

Added

- **Bilingual Documentation Across All Technical Documents:**
 - Full English translations for all technical documentation under `docs/` (`MANUAL.en.md` , `ARCHITECTURE.en.md` , `COMMISSIONING.en.md` , `CONTRIBUTING.en.md` , `DESIGN_SPECIFICATION.en.md` , `STANDARD_DEVELOPMENT_PLAN.en.md` , `TROUBLESHOOTING.en.md` , `CHANGELOG.en.md`).
 - Seamless language toggle links (`English | 日本語`) at the top of every document.
 - Comprehensive English code comments, XML docstrings, and test descriptions across all C# source files.

Changed

- **100% English Standardized UI:**
 - Unified all user interface components, buttons, tabs, filter boxes, tooltips, dialogs, and regex hints into standardized English.
- **License & Copyright Information Update:**
 - Updated `LICENSE` copyright period to `2019-2026 Takashi Kodama` .
 - Added `THIRD-PARTY-NOTICES.md` detailing open-source licenses and notices for bundled packages.

[3.0.0] - 2026-09-26

Added

- **"Dose" Column in Search Results Table:** New column displaying ✓ / — indicating whether 3D dose has been calculated.
- **Version Number in Release Artifacts:** Renamed standalone binary in `release/` to `EclipseDataMiner_v3.0.0.exe` .
- **Automatic PDF Manual Release Synchronization:** Auto-syncs PDF documentation to `release/` during `test.bat` .

Changed

- **Major Version 3.0.0 Release:** Unified AssemblyVersion, InformationalVersion, UI title, and documentation to v3.0.0.

[2.4.0] - 2026-09-25**Added**

- **Advanced Filters for Beam Parameters & Date Range:**
 - Machine ID, Energy, Technique filters supporting comma-separated OR and NOT (`!` , `-`) exclusions.
 - Date range filtering (`TreatmentApprovalDate` , `PlanningApprovalDate` , `CreationDate`).
- **Automated Startup Smoke Testing Pipeline & XAML Resource Integrity Guard:**
 - Automated smoke test launching the release binary post-build in `test.bat` .
 - Automated XAML `StaticResource` integrity validation unit test.
- **Folder-Based Search Preset System (`Presets/` Directory):**
 - App-relative `Presets/` folder storage for seamless sharing and backup.
 - In-place description editing and clinical notes persistence.
- **Numerical Range & Inequality Filtering:**
 - Range (`70-80` , `70~80`) and inequality (`≥ 10` , `≤ 30`) syntax parser.
- **NOT / Exclusion Filters (`!QA` , `!Test`):**
 - Exclusion syntax for all search text fields.
- **Modern Slate & Ocean Cyan Design System:**
 - Enterprise medical UI theme with optimized contrast and visual hierarchy.